"Credit scores are treated as objective measures of risk, yet one recent NBER study finds they misclassify **41% of consumers**, with errors hitting the most vulnerable hardest: **47% of Subprime and 70% of Near Prime borrowers** are placed in the wrong risk category."

NovaCred

MODEL GOVERNANCE AND DATA AUDIT

GROUP 3

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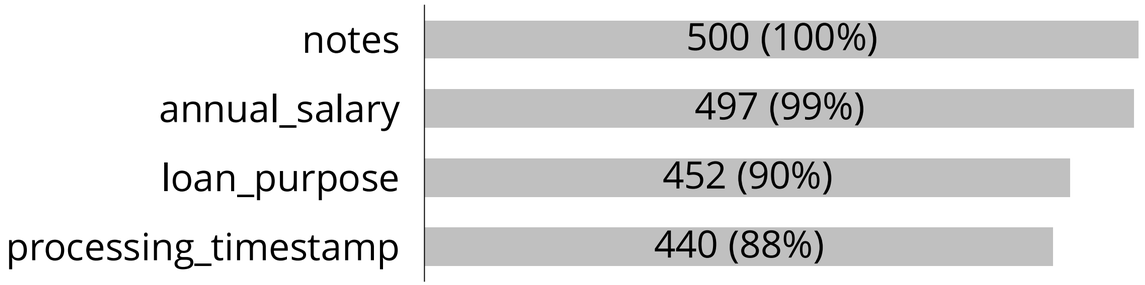
Missing values for decision-critical fields and structural inconsistencies pose moderate risk to Novacred

Completeness

Moderate

Top 10 missing fields (w/o "spending_" fields)



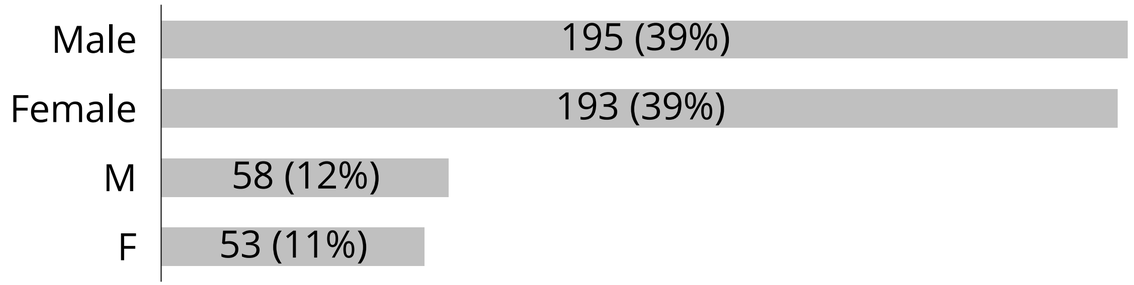
Key Findings

- **>80% of fields have missing values** (>50% "spending_" fields)
- **Auditability and business context** is the main gap: processing timestamp 87.6% and loan purpose 90% missing
- **Schema inconsistency:** annual_income vs. annual_salary
- **Missingness is understated**: empty strings across 14 values
- **Traceability gaps** : ssn and ip_address missing in 1.0% each

Consistency

Moderate

Exp.: Inconsistent gender encoding (3 records NaN)



Key Findings

- **Primary key integrity breach**: 2 duplicated IDs (4 records)
- **Data type inconsistency** annual_income (488 int, 8 str, 1 float)
- **Duplicate-record divergence:** minor within-ID inconsistencies (e.g., SSN missing in one duplicate)
- **Non-standard date formats:** 3 coexisting date_of_birth formats (340 + 101 + 56 records)

Validity

Accuracy

Invalid values and formats and inaccuracies in decision-critical variables pose moderate-high risk to Novacred

Completeness

1 Invalid values

- **Domain violation in DTI:** 1 record (0.2%) with 1.85
- **Implausible values for decision-critical fields:** 2 records (0.4%) with <0 credit history and 1 (0.2%) record with <0 savings balance[1]

2 Invalid formats

- **Invalid contact data:** 1 record (0.2%) with malformed email (missing @)
- **Schema/type mismatches:** Misaligned fields processing timestamp, annual income (object vs. datetime, object vs. float)

Validity — Moderate

Consistency

1 Approved loans w/o or 0 income

- **Decision integrity breach:** 3 approved loans (0.6%) have zero or missing income (due to schema mismatch identified in Completeness)

2 Total spending exceeding reported income

- **Income–spending inconsistency:** 1 record (0.2%) reports income = 0 but total spending = 1,389

Accuracy — Moderate - High

Bias analysis confirms invalid loan approval rates by gender

Analysis

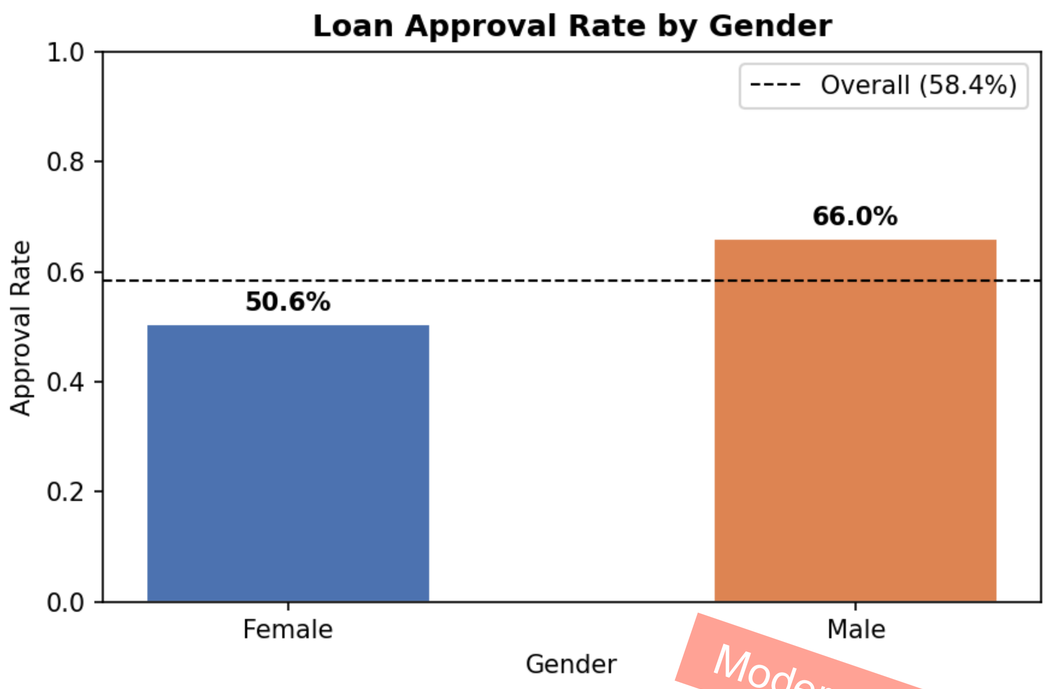
1 Baseline Disparity Analysis

- Approval Rate for females lies 15,4 Percent points below the one for males
- Disparate Impact Ratio is 0.77 below four-fifths threshold of 0.8

2 Conditional Fiarness Analysis

- **Odds ratio** for male is at **1.98** at ceteris paribus of financial numbers

Regulations: Discriminatory Decisioning under GDPR Article 22 and EU AI Act Annex III.



Moderate - High

Deeper Analysis confirms further bias risks through proxy variables

Key Bias Findings with Conclusion

Intersectional Effects (Age x Gender)	High	Female 26–35: DI = 0.62; Female 18–25: DI = 0.77; Female 51–65: DI = 0.76	Overall DI of 0.77 understates subgroup harm — all three violate four-fifths rule
ZIP Code Proxy	High	NYC 88.8% male · LA 93.5% female; $\chi^2 = 324.67$, $p < 0.001$	Not an active proxy (OR = 1.14, p = 0.67) but encodes gender — GDPR Art. 5(1)(c) breach
Age Discrimination	Moderate	Approval: 43.5% (18–25) → 67.4% (36–50) · KW p = 0.003	Not independent after controls (OR = 1.00, p = 0.72) — reflects risk profile differences
Sensitivity Spending Data	Moderate	Gambling, adult entertainment & alcohol categories collected Male	No demonstrated credit-decision necessity — likely violates GDPR Art. 9
Pricing Fairness	Low	4.63% vs. Female 4.49%; Δ = +0.14 pp · OLS p = 0.288	No statistically significant pricing discrimination detected

PII direct and quasi identifiers with almost 100% coverage and (conditional) sensitive categories pose governance risks

PII Identifiers

full_name	100%	**Direct identifier**	Critical
email	98%	**Direct identifier**	Critical
ssn	99%	**Direct identifier**	Critical
ip_address	99%	**Direct identifier**	Critical
date_of_birth[1]	100%	**Quasi identifier**	High
zip_code[1]	100%	**Quasi identifier**	High
gender[1]	100%	**Protected Attr.**	High
sensit. spend. cat.	5%	**Cond. sensitive**	High
loan purp. = medical	2%	**Cond. sensitive**	High

GDPR Mapping

- **Purpose limitation / Data minimisation (Art. 5(1)(b), 5(1)(c))**
  sensitive behavioural fields lack clear necessity
- **Lawfulness / Transparency / Accountability (Art. 6, 13, 5(2))**
  lawful basis and notice traceability are not evidenced
- **Automated decision safeguards (Art. 22)**
  rejection explanations are largely opaque
- **Privacy by design / Confidentiality (Art. 25, 5(1)(f))**
  direct identifiers remain in analytics data
- **Storage limitation / Accountability (Art. 5(1)(e), 5(2))**
  retention and deletion controls are not evidenced

RECOMMENDATIONS

Prioritizing the most severe issues, we recommend 4 legally required measures for Novacred

Before deployment (legally required)

1

Suspend automated approvals and remediate gender bias (DI = 0.77, OR = 1.98 after controls)
No deployment until DI exceeds 0.80

2

Complete DPIA under GDPR Art. 35 with necessity assessment, risk assessment, and documented safeguards.

3

Enforce privacy by default
separate identity store, replace opaque rejections with contestable reason codes, implement consent mechanism with versioning and withdrawal handling

4

Implement EU AI Act Art. 9-15:
Incl. risk management system, audit trail with timestamps and decision logging (Art. 12), transparency documentation (Art. 13), human oversight (Art. 14)

After deployment

- **Ongoing fairness monitoring** with escalation thresholds (DI < 0.85 warning, DI < 0.80 mandatory review).
- **Data retention policy** with automated deletion, deletion audit logs, and DSAR workflow.
- **Governance framework**
  AI system owner, release gates, periodic compliance review.

CONCLUSION

Novacred cannot deploy before properly introducing governance, quality and bias reduction mechanisms

1 Data Quality

16 issues identified

Overall risk: Moderate - High

2 Data Bias

0.77 = DI

1.98 = OR

Overall risk: High

3 Privacy and Governance

Direct identifiers unprotexted

6/7 AI Act obligations unmet

Overall risk: Critical